

Math/Stat 452: Mathematical Statistics

Instructor: Benjamin Ridenhour

Spring, 2023

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Class Hours: MWF 8:30–9:20am
Class Room: TLC 223

Course Description

In this course, we will focus on understanding the interplay between probability, statistics, and modeling. In many regards, probability and statistics are complementary branches of applied mathematics. Probability expresses the chance of observing an outcome given a model (e.g., a distribution); conversely, statistics asks whether a given set of observations are consistent with a parameterization of a chosen mathematical model given an assumed probability distribution. More specifically, in statistics we often seek to find a range of parameter values in which we are confident or to distinguish between competing models with some degree of certainty. Likewise, “inverse problems” in mathematics attempt to find valid parameters given a generative model and observed data, which is simply another way to discuss the practice of statistical model fitting. This course will help you understand and implement the tools required to build complex models and fit them to real-world data.

Course Structure

Our class is intended to be an in-person class, and, as such, content delivery will be done via a live lecture. Obviously, students will not always be able to attend class; I will do my best to accommodate situations as they arise. I recommend you attend if at all possible. Similarly, evaluation (exams, quizzes) are being done in-person (see below).

Coursework will be as follows. Suggested homework problems will be given for each section of the book. Problem sets will be posted to Canvas. Answers to the suggested problems can be found in the back of the book. Friday class periods will be used for quizzes. The graded portion of the class will be your performance on three equally-weighted, non-cumulative exams, in addition to the Friday quizzes. I expect to give 9 quizzes of which you can drop one (so 8 quizzes count toward your grade). Quizzes will cumulatively count as one exam. In other words, each evaluated facet of class is worth 25% of your overall grade. Chapters of the textbook are fairly short. Exams will roughly cover about 9 chapters. *You will be allowed to have a page of notes*

during each exam, but no calculators are allowed! If you cannot be in attendance for a quiz or exam, arrangements need to be made with me prior to the event.

Required Materials

- *Probability and Mathematical Statistics*. By MC Meyer. Society for Industrial and Applied Mathematics: Philadelphia.

Prerequisites/Corequisites

Prerequisites: Math/Stat 451

Topics By Week (*Approximate*)

1. Wks. 1–2: Introduction and review of distributions (discrete, continuous, multivariate); *No class MLK day 16 Jan.*
2. Wk 3: Chs. 42 (Parameter Estimation), 43 (MLE)
3. Wk 4: Chs. 43 (MLE cont.), 44 (MM; W&F)
4. Wk 5: Review, **Test 1**, Ch. 46 (Consistency)
5. Wk 6: Chs. 50 (CIs), 51 (Bootstrapping), 52 (Information)
6. Wk 7: Chs. 53 (Sufficient Statistics), 54 (UMVUE); *No class President's day, 20 Feb.*
7. Wk 8: Ch. 55 (Exponential Family), Numerical methods, Review
8. Wk 9: **Test 2**, Test 2 Summary, *No class 10 Mar.*
9. Wk 10: *Spring Break 13–17 Mar.*
10. Wk 11: Chs. 34 (Sampling a Normal), 35 (Hypothesis Testing), 36 (Hypothesis Testing)
11. Wk 12: Chs. 56 (Size and Power), 57 (Neyman-Pearson Lemma), 58 (LRT)
12. Wk 13: Chs. 58 (LRT cont.), CIs and testing, Review
13. Wk 14: **Test 3**, Ch. 59 (χ^2 -tests; W&F)
14. Wk 15: Models relying on normal distributions (M&W), Ch. 60 (ANOVA)
15. Wk 16: Chs. 45 (Bayesian Estimation), 49 (Credible Intervals; W&F)
16. Wk 17: Model selection, Hierarchical models, MCMC

Course Objectives/Learning Outcomes

Successful students will:

1. Understand the theory of statistics sufficiently to solve most problems appearing in standard textbooks for this course.

Final Exam Date

11 May (Thursday), 8:00–10:00am

Grading Policy

A typical grading scale will be used with cutoffs every 10% (i.e., $\geq 90\%$ = A, $(90\%, 80\%]$ = B, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 25% of your grade will be determined by each of the 3 exams.
- 25% of your grade will be determined by your quiz scores.
 - Calculated using the best 8 out of 9 quizzes.